

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2021/2022

MAY EXAMINATION

UCCD2063 ARTIFICIAL INTELLIGENCE TECHNIQUES

THURSDAY, 19 MAY 2022

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)

Instructions to Candidates :

This question paper consists of FIVE (5) questions.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**. Each question carries 25 marks.

Should a candidate answers more than ONE (1) question in section B, marks will only be awarded for the FIRST ONE (1) question in that section in the order the candidate submits the answers.

Candidates are allowed to use a calculator.

Answer questions only in the answer booklet provided.

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Section A (Answer All Questions)

- Q1. (a) Team A plans to make an attempt to reach the summit of Mount K1 today. The weather forecast predicted that there is only 20% chance of bad weather. From past records, if the weather is bad, only 30% of the teams managed to reach the summit, but if the weather is not bad, 90% of the teams reached the summit. Let S be the event that team A makes it to the summit and let B be the event that the weather is bad.

Some of the required formulae are available in the Appendix.

- Compute the following probabilities: $P(B)$, $P(S|B)$, $P(S|\neg B)$, $P(\neg S|B)$ and $P(\neg S|\neg B)$. (5 marks)
- What is the probability that there is a storm and team A makes it to the summit? (2 marks)
- What is the probability of team A reaching the summit? (3 marks)
- If team A does not make it to the summit, what is the probability that the weather is difficult? (2 marks)

- (b) Figure 1.1 shows a Bayesian Network:

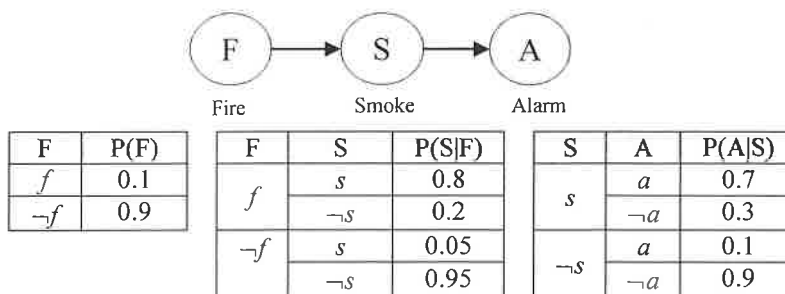


Figure 1.1: Bayesian Network

- Write the equation for the joint distribution $P(F, S, A)$. (3 marks)
- Identify the conditional independence relationship in the network. (2 marks)
- Compute the following probabilities: (8 marks)
 - $P(f, s, a)$
 - $P(s, a)$
 - $P(a | f, s)$

[Total : 25 marks]

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- Q2. The following table shows the distance (km) to travel from a particular town to the other. For cases that two towns are not interlinked or there is one-way direction, their distance is marked as "-". For example, the distance from town A to town D is 5 but there is no road going from A to C, E, or F directly (not interlinked), and no road from D back to A (A to D is one-way).

Distance (km)		Town (to)					
		A	B	C	D	E	F
Town (from)	A	0	6	-	5	-	-
	B	6	0	7	-	-	-
	C	-	-	0	-	-	3
	D	-	-	4	0	2	-
	E	-	-	-	-	0	3
	F	-	-	-	-	-	0

- (a) Draw the state space graph to represent all the interconnections between towns. (4 marks)
- (b) If **Uniform Cost Search (UCS)** is used to find the optimal path to go from town A to F,
- (i) Show the tree search steps in a 2-column table to indicate the *selected node* and the *frontier* list. If there exist nodes with the similar cost, insert node based on alphabetical order. (10 marks)
- (ii) What is the optimal path and cost to go from A to F? (2 marks)
- (iii) What is the memory consumption to run UCS? (2 marks)
- (c) If a **Depth First Search (DFS)** tree based on alphabetical order is used to find the path to go from town A to F, is the solution complete? Justify your answer by drawing the effective search tree (partial tree if solution is incomplete) generated by DFS. (7 marks)
- [Total : 25 marks]
- Q3. (a) Confusion matrix is a better way to evaluate the performance of a classifier. Briefly describe the following terminology in confusion matrix. Some of the required formulae are available in the Appendix.

- (i) True Negative (TN) (2 marks)
- (ii) False Positive (FP) (2 marks)
- (iii) False Negative (FN) (2 marks)

UCCD2063 ARTIFICIAL INTELLIGENCE TECHNIQUES**Q3. (a) (Continued)**

- (iv) True Positive (TP) (2 marks)
- (b) Draw the confusion matrix where each row represents an actual class, while each column represents a predicted class. Fill up each cell in the confusion matrix with the terminology listed in Q3(a). (4 marks)
- (c) Table 3.1 shows the results of a binary classifier.

Table 3.1: Result of Binary Classifier

Actual	Predicted
True	True
False	True
True	False
True	True
True	True
True	False
False	False
False	True
True	True
True	False

Determine the following performance measures for the classifier in Table 3.1.

- (i) Accuracy (2 marks)
- (ii) Precision (2 marks)
- (iii) Recall (2 marks)
- (iv) F_1 score (2 marks)
- (d) Justify the statement that accuracy may not be the suitable performance measure for classifiers for some datasets. (1 mark)
- (e) Binary classifiers can be used in multiclass classification by using the following strategies. Identify ONE advantage and ONE disadvantage for each of the strategies.
- (i) One-versus-All (OvA) (2 marks)
- (ii) One-versus-One (OvO) (2 marks)

[Total : 25 marks]

UCCD2063 ARTIFICIAL INTELLIGENCE TECHNIQUES**Section B (Choose Any ONE (1) Question)**

- Q4. (a) Classify machine learning systems into FOUR (4) different categories based on the *amount of human supervision* involved during the training stage. Then, provide a brief description for each of the category. (12 marks)
- (b) In preparing the data for machine learning algorithms, the promising attributes which are linearly correlated with the output can be found by using the Pearson's correlation coefficient as shown below.

$$r = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{\sqrt{n \sum x_i^2 - (\sum x_i)^2} \sqrt{n \sum y_i^2 - (\sum y_i)^2}}$$

Table 4.1 shows the level of humidity (%) and amount of rainfall (mm) in Kuala Lumpur for the first half of 2017.

Table 4.1 Level of humidity and rainfall in Kuala Lumpur

Month in Year 2017	Humidity (%)	Rainfall (mm)
January	82	341.8
February	81	238.5
March	82	463.1
April	81	536.7
May	80	609.9
June	75	323.1

- (i) Determine the Pearson's correlation coefficient, r , between the level of humidity (%) and amount of rainfall (mm). (8 marks)
- (ii) Provide your comment on the relationship between the level of humidity (%) and amount of rainfall (mm) based on the calculated Pearson's correlation coefficient, r , in (b)(i). (2 marks)
- (c) Feature scaling can transform the raw input features into a representation that is more suitable for the Machine Learning algorithm. List down THREE (3) normalization strategies to transform the raw input features to the same scale. (3 marks)

[Total : 25 marks]

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- Q5. Table 5.1 shows samples of humidity and rainfall collected during different time of the year. The task is to predict the amount of rainfall based on humidity data. Some of the required formulae are available in the Appendix.

Table 5.1 Predicting rainfall based on humidity

Humidity (%)	Rainfall (mm)
81	609.9
80	536.7
75	323.1
71	237.4
69	266.3
78	364.6

- (a) Assume a **linear model** is used to model the problem.
- (i) Given the two models $\Theta_A = (-1700, 27)$ and $\Theta_B = (200, 10)$. Compute the MSE for both models. Which model fits the samples better? (9 marks)
- (ii) **Gradient descent** is used to find the best parameter to model the data. Given that the parameter values is initialized to Θ_B , compute the gradient descent after the *first* iteration. (7 marks)
- (b) Rather than predicting the rainfall amount, we want to build a system classify the rainfall into two categories (LOW and HIGH) based on their amount.
- (i) Convert the samples in Table 5.1 into the two categories using **equal-distance partitioning**. (6 marks)
- (ii) Logistic Regression and Neural Network are two types of classification algorithm. Explain why is it more difficult to learn the parameters of a neural network than a logistic regression classifier? (3 marks)
- [Total : 25 marks]

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$$Accuracy = \frac{TN + TP}{TN + FP + FN + TP}$$

$$Precision = \frac{TP}{FP + TP}$$

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

$$F_1 = \frac{TP}{TP + \frac{FN + FP}{2}}$$

$$Recall = \frac{TP}{FN + TP}$$

$$h(\mathbf{x}) = \theta^T \mathbf{x}$$

$$\mathbf{h} = \mathbf{X} \cdot \theta$$

$$MSE(\theta) = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

$$RMSE(\theta) = \sqrt{MSE(\theta)}$$

$$\nabla_{\theta} MSE(\theta) = \frac{2}{m} \mathbf{X}^T \cdot (\mathbf{X} \cdot \theta - \mathbf{y})$$

$$P(A, B) = P(A|B)P(B)$$

$$P(A, B) = P(B|A)P(A)$$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$P(A) + P(\neg A) = 1$$

